

CYBER SECURITY & CYBER CRIMES

Types of Cyber Threats

Preventive Measures

Real-Life Case Studies

Prepared by : Yajas Mishra
Class : IX - Excellent

// SUMMER VACATION PROJECT

TABLE OF CONTENTS

Index

01

Introduction to Cyber Security

pg. 03

03

Types of Cyber Threats

pg. 05

05

Preventive Measures

pg. 07

07

Conclusion

pg. 11

02

What is Cybercrime?

pg. 04

04

Common Cyber Attack Methods

pg. 06

06

Real-Life Case Studies

pg. 08

08

Bibliography

pg. 12

What is Cyber Security?

Cyber security is the practice of protecting computers, networks, programs, and data from unauthorised access, damage, or attack. As the internet connects billions of people, securing digital systems has become one of the most critical challenges of our time.

The CIA Triad – Core Principles



CONFIDENTIALITY

Data is accessible only to those who are authorised to see it.



INTEGRITY

Data remains accurate and has not been tampered with.



AVAILABILITY

Data and systems are accessible when needed by authorised users.

Why It Matters:

- Protects personal & financial data
- Prevents identity theft & fraud
- Secures national infrastructure
- Ensures safe internet for everyone

What is Cybercrime?

Cybercrime refers to any criminal activity that involves a computer, network, or the internet. Cybercriminals use technology to steal information, disrupt services, extort money, or harm individuals and organisations. In India, the IT Act 2000 makes these offences punishable by law.

Common Targets:



Individuals

Personal data,
money,
photos stolen



Businesses

Trade secrets,
customer data
leaked



Hospitals

Patient records
encrypted,
disrupted



Government

Espionage,
system disruption



Banks

Fraudulent
transfers, ATM
heists



IT Act 2000

India's primary
law against
cybercrime.
Sections 66–74
cover hacking,
identity theft,
data theft, and
more.

Types of Cyber Threats



Malware

Harmful software — viruses, worms, trojans, spyware



Phishing

Fake emails/messages to steal passwords or money



Ransomware

Encrypts your files, demands payment to restore them



Hacking

Unauthorised access to steal, alter, or delete data



Identity Theft

Stealing personal info to commit fraud in your name



Cyberbullying

Harassing, threatening, or humiliating people online



DDoS Attack

Flooding servers to crash websites and services



Man-in-the-Middle

Intercepting communication between two parties

How Attackers Strike

01

Phishing

- Fake emails/SMS/calls from trusted sources
- Tricks users into clicking malicious links
- Types: Spear, Smishing, Vishing

02

Malware & Ransomware

- Malware = viruses, worms, trojans, spyware
- Ransomware encrypts files, demands Bitcoin
- Spread via downloads, email attachments

03

Social Engineering

- Exploits human psychology, not tech
- Tactics: pretexting, baiting, tailgating
- Most dangerous – no software can stop it

04

Password Attacks

- Brute Force – tries every combination
- Dictionary Attack – uses common words
- Credential Stuffing – uses leaked passwords

How to Stay Safe Online

FOR INDIVIDUALS

 Use strong, unique passwords (12+ chars)

 Enable Two-Factor Authentication (2FA)

 Never click suspicious links or attachments

 Keep your software and antivirus updated

 Avoid public Wi-Fi for banking or login

 Back up your data regularly

GOVERNMENT & LEGAL



IT Act 2000

Primary Indian law against cybercrime – Sections 66-74



CERT-In

India's national cyber emergency response team



cybercrime.gov.in

Report cyber crimes online on this portal



2FA Everywhere

Most effective single step to secure accounts



AIIMS Delhi Ransomware Attack

2022

DATE

November 23, 2022

TARGET

AIIMS New Delhi (India's top govt. hospital)

ATTACK TYPE

Ransomware

RECORDS HIT

~4 Crore patient records compromised

IMPACT

Hospital ran on paper for nearly 2 weeks

RANSOM ASKED

30 Bitcoin ≈ ₹1.5 Crore

WHAT HAPPENED

Hackers encrypted all servers of AIIMS Delhi – patient records, appointment systems, and hospital management software became completely inaccessible. Emergency, outpatient, and inpatient services were severely disrupted. Thousands of patients faced delays in treatment.

💡 LESSON LEARNED

Critical institutions must have offline backups, strict access controls, and network segmentation. Healthcare data is an extremely high-value target for hackers.



WannaCry Global Ransomware Attack

2017**DATE**

May 12, 2017

SCALE

200,000+ computers across 150+ countries

EXPLOIT

EternalBlue – Windows vulnerability

UK NHS LOSS

£92 million – operations cancelled

GLOBAL COST

USD 4-8 Billion

RANSOM

USD 300-600 per victim (Bitcoin)

WHAT HAPPENED

WannaCry exploited an unpatched Windows vulnerability (EternalBlue) to spread automatically across networks. It encrypted files and demanded Bitcoin ransom. UK's NHS had to cancel thousands of medical appointments and operations. The vulnerability had been patched by Microsoft two months earlier – but many organisations had not updated.

💡 LESSON LEARNED

Always apply software updates and security patches immediately. Most major attacks exploit vulnerabilities that already have a fix available.



Pune Cosmos Bank Cyber Heist

2018

DATE

August 11-13, 2018

TARGET

Cosmos Cooperative Bank, Pune

ATTACK TYPE

ATM Switching Hack + SWIFT Fraud

AMOUNT STOLEN

₹94.42 Crore (≈ USD 13.5 Million)

COUNTRIES

ATM withdrawals in 28 countries simultaneously

METHOD

450 cloned Visa debit cards used in hours

WHAT HAPPENED

Hackers installed malware on Cosmos Bank's ATM server. They then simultaneously withdrew cash from ATMs in 28 countries using 450 cloned debit cards – all within hours. Separately, fraudulent SWIFT messages transferred money to Hong Kong. One of the largest bank cyber heists in Indian history.

💡 LESSON LEARNED

Banks must monitor unusual transaction patterns in real time. Network segmentation is critical – ATM systems should be isolated from the main banking network.

Conclusion

Cyber security is no longer optional. It is an essential life skill. As more of our lives move online, the threat grows. The good news is that most attacks can be prevented with awareness and simple habits.

Key Takeaways:

- 01 Cyber attacks affect individuals, hospitals, banks, and governments
- 02 Phishing and ransomware are the most common attack types today
- 03 Strong passwords + 2FA can prevent most account breaches
- 04 Always update software – most attacks exploit known, patched bugs
- 05 Every internet user has a role in keeping the digital world safe

"In cyberspace, the first line of defence is an informed user."

Bibliography

01

National Cyber Security Policy – Government of India

cert-in.org.in

03

Cyber Crime Reporting Portal – India

cybercrime.gov.in

05

WannaCry Ransomware – Europol Cyber Crime Report 2017

europol.europa.eu

07

NCERT Class IX Information Technology Textbook

ncert.nic.in

02

Information Technology Act, 2000 – MeitY, India

meity.gov.in

04

AIIMS Ransomware Attack – The Hindu, November 2022

thehindu.com

06

Cosmos Bank Cyber Heist – RBI Cyber Security Report

rbi.org.in

08

What is Cyber Security? – Kaspersky Lab

kaspersky.com

THANK YOU

for your attention

Cyber Security & Cyber Crimes

Presented by: Yajas Mishra
IT-402 | Class IX - Excellent | 2025-26